Argument Distribution

Step 1: Fixed Positional Filling
Step 2: Default Value Overwriting
Step 3: The *args Collector (The "Bucket")
Step 4: The **kwargs Collector (The "Dictionary")

```
def func(a, b=2, *args, **kwargs):
```

a → positional
b → default
*args → unlimited positional tuple
kwargs → dictionary of named args

```
func(10, 20, 3, 4, 5, x=100)
```

a = 10
b = 20
args = (3,4,5)
kwargs = {x:100}

```
func(10, b=5)
```

a = 10
b = 5
args = ( )
kwargs = { }

```
func(a=10, 20)
```

positional arg after named arg

```
def func(a, *args, b=2, **kwargs)
```

```
func(a=10, 20)
```

SyntaxError: positional argument follows keyword argument

```
func(10, a=20)
```

TypeError: func() got multiple values for argument 'a'

```
func(1, 2, 3, 4, b=5)
```

TypeError: func() got multiple values for argument 'b'